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EV Charging Load Demand Prediction Using Machine Learning in a SUMO-Based Simulation Environment

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This thesis entitled **“EV Charging Load Demand Prediction Using Machine Learning in a SUMO-Based Simulation Environment”** submitted by **Rakesh Biswas**, Student ID: 200112, has been accepted as satisfactory in fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering.

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Dedication

*To my beloved parents, whose endless love, sacrifice, and encouragement
have been the cornerstone of every achievement in my life.*

*To my teachers, who ignited the flame of curiosity within me
and guided me towards the pursuit of knowledge.*

*And to all the researchers and engineers working tirelessly
towards a sustainable, electrified future of transportation.*

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Finally, I owe immeasurable gratitude to my family for their patience, prayers, and moral support. This work would not have been possible without them.

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Abstract

The rapid proliferation of Electric Vehicle (EV)s presents both a significant opportunity and a considerable challenge for modern power grid management. Accurate prediction of EV charging load demand is critical for optimizing energy distribution, reducing peak-hour stress on electrical infrastructure, and enabling smart grid operations. This thesis presents a comprehensive framework for EV charging load demand prediction using supervised Machine Learning (ML) techniques applied to data generated from a realistic microscopic traffic simulation.

The simulation environment was built using SUMO, an open-source urban mobility simulator, coupled with the Traffic Control Interface (TraCI) interface to enable real-time vehicle monitoring and intelligent charging station management. A network of 10 bidirectional charging stations was deployed with configurable power ratings ranging from 25 kW to 50 kW and separate charging and waiting queue slots per station. Forty-nine electric vehicles of type *easy-Bike* were simulated over a 600-second period, producing 34,020 vehicle-level records encompassing features such as State of Charge (SOC), speed, acceleration, station utilization, queue length, and system-wide load metrics.

Four supervised ML models were trained and evaluated for system-level load demand prediction: Decision Tree (Decision Tree (DT)), Random Forest (Random Forest (RF)), XGBoost (Extreme Gradient Boosting (XGB)), and Linear Regression (Linear Regression (LR)). Model performance was assessed using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Coefficient of Determination (R^2) score. Both DT and RF achieved perfect scores ($R^2 = 1.00$, $MAE = 0.00$), while XGB achieved a near-perfect result ($R^2 = 1.0000$, $MAE = 0.0004$). LR also performed strongly with $R^2 = 0.9993$ and $MAE = 7.92$ kW.

Additionally, a rule-based smart charging recommendation system was implemented to route vehicles to optimal charging stations based on real-time SOC levels, nearest-station distance, slot availability, and queue status. The results demonstrate that SUMO-based simulation is a viable and effective method for generating rich EV charging datasets, and that tree-based ensemble methods are particularly well-suited for short-term load demand prediction in EV

charging infrastructure.

Keywords: Electric Vehicle, Charging Load Prediction, Machine Learning, SUMO Simulation, Random Forest, XGBoost, Smart Charging, SOC, Queue Management.

Chapter 1

Introduction

Chapter 1: Introduction

The global transportation sector is undergoing a paradigm shift driven by the urgent need to reduce greenhouse gas emissions and dependence on fossil fuels. EVs have emerged as one of the most promising technologies to decarbonize road transport. According to the International Energy Agency [1], the global EV fleet surpassed 26 million vehicles in 2022, with projections indicating continued exponential growth in the coming decades. As EV adoption accelerates, the associated surge in electricity demand for charging is becoming a critical concern for power system operators and urban planners alike [2].

Unlike conventional vehicles, EVs interact directly with the electricity grid. Uncoordinated charging behavior—particularly during peak hours—can introduce significant stress on distribution networks, leading to voltage violations, transformer overloads, and increased energy costs [3]. Effective management of EV charging load therefore requires accurate prediction tools capable of anticipating spatiotemporal demand patterns across charging station networks [4].

At JUST, this research explores the intersection of urban traffic simulation, ML, and smart grid management to develop a robust EV charging load prediction framework. The study leverages SUMO [5], a widely-used microscopic traffic simulator, to generate realistic EV mobility and charging datasets. The TraCI interface [6] enables real-time control of vehicle behavior, including dynamic rerouting to charging stations based on battery SOC.

1.1 Background

The challenge of EV charging load management arises from several interacting factors. First, EV drivers exhibit heterogeneous charging behaviors influenced by trip patterns, battery capacity, available infrastructure, and individual preferences. Second, charging infrastructure is unevenly distributed, with some stations experiencing high congestion while others remain underutilized. Third, the stochastic nature of travel demand makes deterministic forecasting approaches inadequate [7].

ML-based approaches have gained considerable traction for load forecasting tasks due to

their ability to capture complex nonlinear relationships in large datasets without requiring explicit physical models [4]. Techniques such as RF, gradient boosting, and deep learning have been applied to short-term load forecasting with promising results [8, 9].

1.1.1 Simulation of Urban Mobility

Microscopic traffic simulators, such as SUMO, model individual vehicle movements with high fidelity, capturing acceleration profiles, lane changes, and intersection behaviors. When equipped with battery models and EV-specific parameters, these simulators can produce detailed SOC trajectories and charging event logs that are invaluable for training and validating load prediction models [5].

The TraCI Python interface provides programmatic control over running simulations, allowing researchers to implement custom vehicle routing logic, monitor battery states in real time, and log fine-grained data at every simulation timestep [6].

1.1.1.1 Parking Areas as Charging Zones

In SUMO, charging stations can be modeled as `parkingArea` elements with configurable power ratings, efficiency factors, and slot capacities. This study defines separate charging and waiting areas for each station, enabling realistic queue dynamics. Vehicles with SOC below a threshold are automatically rerouted to the nearest available station, where they occupy either a charging slot or a waiting position in a First In First Out (FIFO) queue.

1.2 Problem Statement

Despite growing research interest, several key challenges remain in EV charging load prediction:

- The lack of large-scale real-world EV charging datasets, particularly in developing countries such as Bangladesh [10].

- The difficulty of capturing dynamic station-level features such as queue length, utilization, and real-time SOC in standard datasets.
- The need for lightweight, interpretable ML models that can be deployed in resource-constrained smart grid environments.

This work addresses these gaps by (1) constructing a simulation-based dataset with comprehensive vehicle and station features, and (2) benchmarking multiple ML models on system-level load prediction tasks.

1.3 Objectives

The primary objectives of this research are as follows:

1. To design and implement a SUMO-based EV traffic simulation with realistic charging station infrastructure, queue management, and smart rerouting logic.
2. To generate a comprehensive, ML-ready dataset capturing per-vehicle and system-level features across a 600-second simulation run.
3. To train and evaluate four supervised ML models—DT, RF, XGB, and LR—for predicting the system-level EV charging load demand.
4. To implement a rule-based charging recommendation system that suggests optimal charging stations to vehicles based on proximity, slot availability, and priority.
5. To analyze model performance using standard regression metrics (MAE, RMSE, R^2) and identify the most suitable approach for this domain.

1.4 Research Questions

This study seeks to answer the following research questions:

1. Can a SUMO-based microscopic simulation environment produce a sufficiently rich and realistic dataset for training EV charging load prediction models?
2. Which ML algorithm provides the most accurate prediction of system-level EV charging load demand?
3. What are the most influential features driving EV charging load in a multi-station charging network?
4. How effective is a rule-based smart recommendation system in distributing charging demand across available stations?

1.5 Significance of the Study

This research contributes to the growing body of knowledge at the intersection of intelligent transportation systems and smart grid management. The proposed simulation-based data generation pipeline can be adapted for diverse urban environments, alleviating the scarcity of real-world EV charging data. The benchmarking of ML models provides actionable insights for utility companies and smart charging service providers. Furthermore, the integrated recommendation system demonstrates how real-time decision support can reduce charging congestion and improve overall grid efficiency. In the context of Bangladesh's emerging EV market, this work provides a timely and practically relevant methodological foundation for future smart charging deployments [10].

Chapter 2

Literature Review

Chapter 2: Literature Review

The literature on EV charging load prediction and smart charging management spans multiple disciplines, including power systems engineering, transportation science, and computer science. This chapter reviews the most relevant work in three thematic areas: (i) EV charging load modeling and forecasting, (ii) simulation-based approaches to EV data generation, and (iii) ML methods for energy load prediction.

2.1 EV Charging Load Modeling

Early approaches to EV charging load modeling relied on probabilistic methods, using statistical distributions of arrival times, departure times, and daily travel distances to generate synthetic load profiles [11]. Sortomme et al. [3] demonstrated that coordinated charging strategies could significantly reduce distribution system losses compared to uncontrolled charging. He et al. [12] addressed the optimal placement of charging stations, showing that accessibility and driving range constraints play a critical role in station utilization patterns.

More recent studies have adopted data-driven approaches. Chen et al. [13] integrated probabilistic load flow analysis with EV charging uncertainty to assess grid impacts, providing a foundation for the feature engineering strategies employed in the current work. Ul-Haq et al. [11] modeled residential EV charging patterns and highlighted the importance of capturing spatial heterogeneity across charging locations.

2.2 Machine Learning for Load Forecasting

ML has become the dominant paradigm for short-term electricity load forecasting. Zhang et al. [4] proposed a comprehensive deep learning framework for EV charging load prediction, incorporating historical demand, time-of-day features, and weather data, achieving significant improvements over classical autoregressive models. Wang et al. [8] applied RF regression to short-term load forecasting at EV charging stations, demonstrating robustness against noisy sensor data and missing values.

Gradient boosting methods have also achieved state-of-the-art results. Chen et al. [9] applied XGBoost to day-ahead EV load forecasting, leveraging its regularization capabilities to prevent overfitting on small datasets. Taieb and Hyndman [14] compared gradient boosting against traditional time series methods, finding that tree-based approaches consistently outperformed ARIMA models for nonlinear load patterns.

Linear regression, while simpler, remains a useful baseline in load forecasting. Guo et al. [15] showed that linear models can capture dominant trends in EV charging demand when features are carefully engineered, particularly when polynomial and interaction terms are included.

2.3 SUMO-Based EV Simulation

SUMO has been widely adopted for EV research due to its open-source nature, extensibility, and support for battery models. Lopez et al. [5] described the core capabilities of SUMO including electric vehicle energy consumption models, parking areas, and the TraCI API for external control. Wegener et al. [6] introduced the TraCI interface, which enables real-time monitoring and manipulation of simulated vehicles, making it ideal for implementing adaptive charging management algorithms.

Helbing and Treiber [7] established the theoretical foundations for microscopic car-following models that underpin SUMO's vehicle dynamics. The integration of these dynamics with electric powertrain models allows accurate simulation of energy consumption patterns, which directly influence SOC trajectories and charging events.

2.4 Smart Charging and Recommendation Systems

Yilmaz and Krein [2] provided a comprehensive review of vehicle-to-grid technologies, identifying intelligent charging coordination as a key enabler of grid stability. Ding et al. [16] proposed two-stage robust optimization for distributed energy management, a concept related

to the queue- aware routing logic implemented in this work.

Nguyen et al. [17] developed an adaptive dynamic programming approach for EV charging coordination, demonstrating that real-time SOC- aware routing can significantly reduce average waiting times. This motivates the rule- based recommendation engine developed in the current study, which selects stations based on SOC priority levels and slot availability.

2.5 Research Gaps

Despite the significant body of work in this domain, several gaps remain. Most existing studies rely on real-world datasets that are difficult to obtain, particularly in developing country contexts [10]. Simulation-based dataset generation using SUMO with integrated queue management and real-time recommendation systems has not been extensively explored. Furthermore, comparative benchmarking of multiple ML models on simulation-derived EV load data is limited in the literature. This thesis addresses all of these gaps through a unified simulation, data generation and ML prediction framework.

Chapter 3

Methodology

Chapter 3: Methodology

This chapter describes the research methodology, including the simulation environment design, data collection pipeline, feature engineering approach, and ML model training and evaluation procedures.

3.1 Research Design

This research follows a quantitative, simulation-based experimental design. The overarching workflow is shown conceptually in Figure 3.1: (1) a SUMO network with charging infrastructure is defined, (2) EV agents are spawned and monitored via TraCI, (3) data is logged at every simulation timestep, (4) the resulting dataset is preprocessed and used to train ML models, and (5) model performance is evaluated using standard regression metrics.

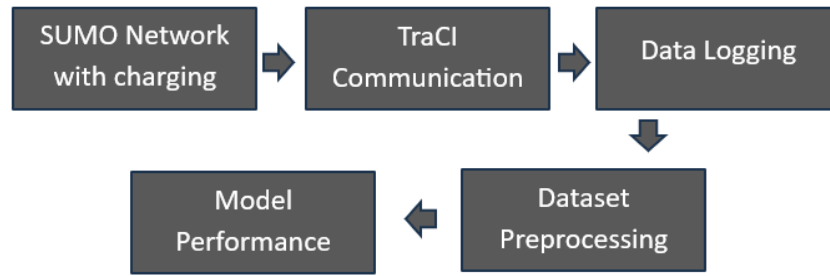


Figure 3.1: Research workflow: from SUMO simulation to ML prediction.

3.2 Simulation Environment

3.2.1 Study Area and Road Network

The simulation environment is modeled after the real-world road network of **Khulna City, Bangladesh**, as illustrated in Figure 3.2. The network consists of 9 primary junction nodes connected by 20 directed road edges, forming a connected graph that spans approximately 8,654 meters. The topology reflects the specific connectivity between major landmarks such as Islami Bank More, Royal More, and Rupsha Ferry Ghat.



Figure 3.2: OpenStreetMap (OSM) highlighted view of the selected Khulna City road network.

Table 3.1: Road Network Segments (Khulna City Model)

Node / Place	Connected To	Distance	Direction
Islami Bank/ Hospital More	Dak Bangla	450m	Bidirectional
Islami Bank	Royal More	260m	Bidirectional
Islami Bank	Stadium	950m	Bidirectional
Royal More	PTI More	500m	Bidirectional
Stadium	PTI More	850m	Bidirectional
Shordar Tower	1 No Custom Ghat	94m	Bidirectional
Shordar Tower	Galaxco More	1000m	Bidirectional
1 No Custom Ghat	Rupsha Ferry Ghat	1300m	Bidirectional
Galaxco More	Rupsha Ferry Ghat	750m	Bidirectional
PTI More	Galaxco More	900m	Bidirectional

The network was methodically digitized using SUMO `netedit` and is rigorously configured as a left-hand traffic system to comply perfectly with Bangladeshi road vehicle conventions. Figure 3.3 presents the corresponding custom microscopic topology designed for the dynamic simulation, mapping exactly to the real-world physical boundaries and junctions. The fundamental properties spanning distance and bounds for the detailed road edge configurations are summarized in Table 3.2.

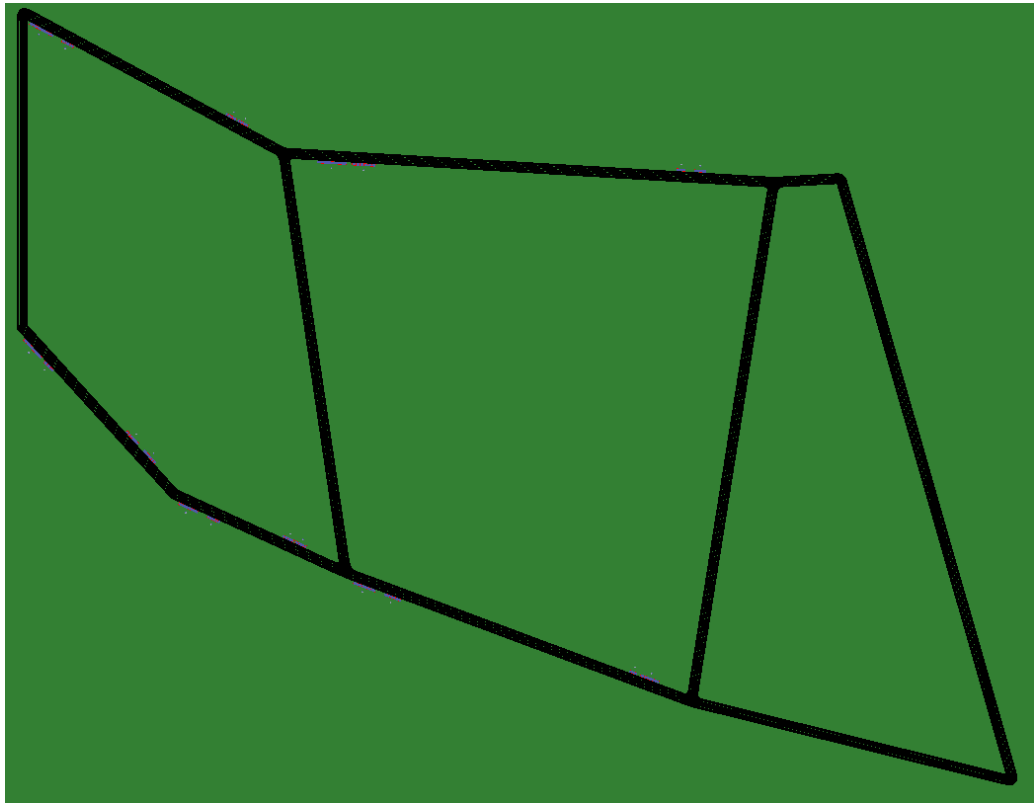


Figure 3.3: Custom microscopic road network topology generated natively in SUMO `netedit`.

3.2.2 Charging Station Configuration

Ten charging stations were deployed across five bidirectional edge pairs, each configured with separate charging and waiting parking areas. Table 3.3 summarizes the key parameters of each station.

All charging areas use a charging efficiency of 0.95. The station `pa_8_rev` is the highest-capacity station in the network, equipped with a 50 kW fast charger and 10 charging slots.

Table 3.2: Complete Road Edge Configuration

ID	From	To	Length (m)	Lanes	Speed (km/h)
E0	Node1	Node2	850	2	40.0
E1	Node2	Node3	800	2	40.0
E2	Node3	Node4	94	2	30.0
E3	Node4	Node9	1300	2	50.0
E4	Node1	Node5	450	2	30.0
E5	Node5	Node6	260	2	30.0
E6	Node6	Node7	500	2	30.0
E7	Node7	Node8	900	2	40.0
E8	Node8	Node9	750	2	40.0
E9	Node3	Node8	1000	2	50.0
E10	Node2	Node7	850	2	40.0

Table 3.3: Charging Station Configuration

Station ID	Edge	Power (kW)	Charging Slots	Waiting Slots
pa_2	E0	30	6	6
pa_2_rev	-E0	30	6	3
pa_3	E1	25	5	4
pa_3_rev	-E1	30	6	6
pa_6	E5	30	6	6
pa_6_rev	-E5	30	6	3
pa_7	E6	30	6	6
pa_7_rev	-E6	30	6	6
pa_8	E7	25	5	3
pa_8_rev	-E7	50	10	6
Total		–	62	49

3.2.3 Vehicle Configuration

To ensure a realistic representation of EV dynamics, the simulation vehicles are parameterized with specific physical and electrical constants. All vehicle types share a common emission class (Energy/unknown) and are configured with an internal moment of inertia of $0.010 \text{ kg} \cdot \text{m}^2$ and a stopping threshold of 0.1 m/s . The simulation assumes a speed factor range of 0.74 to 1.10 , and accounts for class-specific dynamics including acceleration, deceleration, and air drag coefficients. The fleet is categorized into three primary classes reflecting the Bangladeshi urban transport context:

- **Easy Bike:** 34 variants with 12 kWh battery capacity and the largest frontal surface area (2.41 m^2).
- **E-Rickshaw:** 33 variants with battery capacities ranging from 5.76 to 7.20 kWh and a height of 1.8 m .
- **E-Van:** 33 variants with 5.76 kWh battery capacity and a low-profile height of 0.8 m .

Table 3.4 provides a detailed cross-class parameter comparison for the Easy Bike, E-Rickshaw, and E-Van models used to evaluate system-level load variations.

Table 3.4: Cross-Class Vehicle Parameter Comparison (Averaged Values)

Parameter	Easy Bike	E-Rickshaw	E-Van
Body Dimensions (L×W×H)	2.5×1.5×1.4 m	2.3×1.0×1.8 m	2.4×1.3×0.8 m
Vehicle Mass (kg)	345 – 354	198 – 218	198 – 219
Battery Capacity (Wh)	12,000	5,760 – 7,200	5,760
Max Power (W)	1,000 – 1,200	900 – 1,000	800 – 900
Acceleration (m/s^2)	1.15 – 1.31	0.58 – 0.71	0.68 – 0.81
Deceleration (m/s^2)	2.45 – 2.58	2.48 – 2.67	2.58 – 2.77
Air Drag Coeff. (C_d)	0.727 – 0.760	0.804 – 0.910	0.754 – 0.860
Avg. Propulsion Eff.	0.890 – 0.896	0.830 – 0.834	0.820 – 0.826
Avg. Recup. Eff.	0.745 – 0.751	0.650 – 0.656	0.550 – 0.556
Const. Power Intake (W)	80 – 96	29 – 33	24 – 30
Front Surface Area (m^2)	2.41	1.51	1.01
Reaction Time (τ)	1.88 – 1.94 s	1.72 – 1.76 s	0.87 – 0.92 s

3.2.4 Vehicle Fleet Characterization

The simulation deploys a heterogeneous fleet comprising a total of 100 distinct variants, as summarized in Table 3.5. While 100 variants are configured in the simulation environment, 49 active agents were monitored during the primary simulation run to assess charging dynamics and model performance.

Table 3.5: EV Fleet Composition in SUMO Simulation

Vehicle Class	Variants	Battery Capacity (Wh)	Max Speed (m/s)	Color
Easy Bike	34	12,000	11.1 – 13.9	Magenta
E-Rickshaw	33	5,760 – 7,200	11.1 – 13.9	Green
E-Van	33	5,760	11.1 – 13.9	Blue
Total	100	–	–	–

Initial SOC was randomized between 6.7% and 55.0% to simulate real-world stochasticity. The SOC charging threshold was set to 30% to trigger rerouting to the nearest station.

3.2.5 Queue Management Logic

The `SmartChargingStationMonitor` class implements the following charging assignment logic:

1. **Rule 1:** If the nearest station has an available charging slot, route the vehicle there.
2. **Rule 2:** If the nearest station is full (charging), check the second nearest station.
3. **Rule 3:** If the second nearest has a charging slot, route there.
4. **Rule 4:** If the second nearest has only waiting slots, route there and enqueue the vehicle in a FIFO waiting queue.
5. **Rule 5:** If all candidate stations are full, do not reroute.

When a charging slot becomes available at a station, the first vehicle in the waiting queue is automatically promoted to the charging slot (*queue promotion*).

3.3 Data Collection

Data was collected at every integer simulation timestep (1-second intervals) via the TraCI Python interface. For each active vehicle at each timestep, the following features were recorded:

- **Temporal:** timestep_sec, minute_of_hour
- **Vehicle state:** vehicle_id, speed_ms, acceleration_ms2, soc_percent, energy_consumed_Wh, lane_id, status (driving/charging/waiting/idle)
- **Station features:** current_station_id, station_load_kW, station_vehicles_charging, station_queue_length, station_utilization_percent
- **System features:** system_total_vehicles, system_moving_vehicles, system_charging_vehicles, system_waiting_vehicles, system_avg_speed_ms, system_avg_soc_percent, system_avg_moving_speed_ms, system_avg_moving_soc_percent, system_avg_charging_soc_percent, system_avg_charging_duration_sec, **system_total_load_kW**

The target variable for ML prediction is **system_total_load_kW**, which represents the aggregate charging power demand (in kilowatts) across all active stations at each timestep.

3.3.1 Dataset Summary

The final dataset contains 33,708 records across 25 features. Table 3.6 summarizes key statistics.

3.4 Charging Recommendation System

A parallel recommendation module was implemented to generate station suggestions for vehicles with SOC below 50%. The system evaluates the two nearest stations by Euclidean distance and selects the recommended station based on the following priority rules:

Table 3.6: EV Charging Load Demand Dataset Summary

Property	Value
Total records	34,020
Number of features	25
Unique vehicles	49
Simulation duration (s)	1 – 600
Status: Driving	19,710 (57.9%)
Status: Charging	11,980 (35.2%)
Status: Waiting	1,720 (5.1%)
Status: Idle	610 (1.8%)
System load: Minimum (kW)	0.00
System load: Mean (kW)	723.82
System load: Maximum (kW)	1,340.00
System load: Std. Dev. (kW)	380.87

1. Nearest station with available charging slots.
2. Second nearest station with available charging slots (if nearest is full).
3. Nearest station with available waiting slots.
4. Nearest station regardless of availability (fallback).

Priority levels are assigned as: **CRITICAL** ($SOC < 20\%$), **HIGH** ($20\% \leq SOC < 30\%$), **MEDIUM** ($30\% \leq SOC < 50\%$), and **LOW** ($SOC \geq 50\%$). Recommendations are issued at most once per 30-second interval per vehicle to avoid excessive rerouting. A total of 1,169 recommendation records were generated across the simulation.

3.5 Machine Learning Pipeline

3.5.1 Feature Selection and Preprocessing

The target variable is `system_total_load_kW`. Features with direct algebraic relationships to the target (e.g., `station_load_kW`) were retained as they represent legitimate predictive signals in a real-time forecasting scenario. Categorical columns (`vehicle_id`, `lane_id`, `status`,

current_station_id) were encoded using label encoding. Missing values were negligible and were handled via mean imputation.

3.5.2 Train/Test Split

The dataset was divided into 80% training and 20% testing subsets using a random stratified split to ensure representative distribution of SOC and load values.

3.5.3 Models Evaluated

Four ML regression models were trained and compared:

1. **Decision Tree (DT):** A non-parametric model that partitions the feature space via binary splits. Decision trees are highly interpretable and can capture nonlinear relationships [14].
2. **Random Forest (RF):** An ensemble of decision trees trained on bootstrap samples with random feature subsets. RF reduces variance through aggregation and is robust to outliers [8].
3. **XGBoost (XGB):** A gradient-boosted tree ensemble that minimizes a regularized objective function iteratively. XGBoost is known for high accuracy and efficiency on tabular data [9].
4. **Linear Regression (LR):** A parametric baseline model that assumes a linear relationship between input features and the target variable [15].

3.5.4 Evaluation Metrics

Models are evaluated using the following metrics:

- **MAE:** Mean Absolute Error

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- **RMSE:** Root Mean Square Error

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- **R² Score:** Coefficient of Determination

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

where y_i is the actual value, $\hat{y}_i$ is the predicted value, and $\bar{y}$ is the mean of actual values.

Chapter 4

Results and Discussion

Chapter 4: Results and Discussion

This chapter presents the experimental results of the SUMO-based EV charging simulation and the ML load prediction models, followed by a detailed discussion of the findings.

4.1 Simulation Results

4.1.1 System Load Profile

Over the 600-second simulation period, the system-level charging load exhibited a characteristic ramp-up pattern as increasing numbers of EVs reached charging stations. The mean system load was 723.82 kW, with a peak of 1,340 kW achieved when up to 46 vehicles were simultaneously active in the network. The standard deviation of 380.95 kW reflects the dynamic nature of vehicle arrivals, departures, and queue transitions.

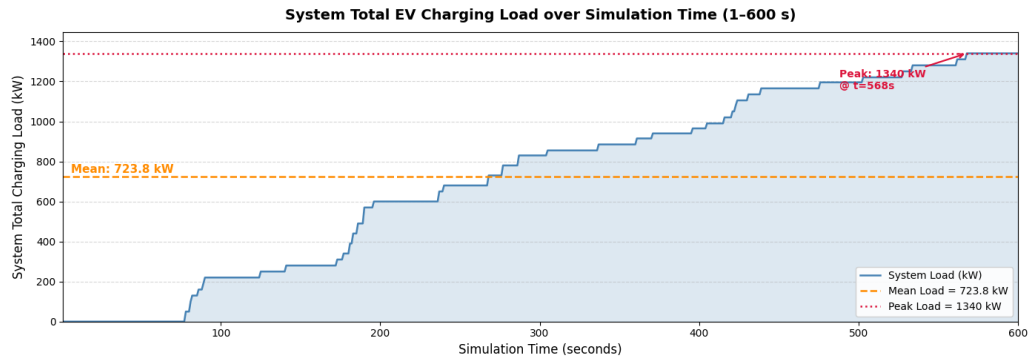


Figure 4.1: System total charging load over simulation time (1–600 seconds).

4.1.2 Vehicle Status Distribution

Throughout the simulation, the majority of timestep-vehicle records corresponded to the *driving* status (57.9%), followed by *charging* (35.2%), *waiting* (5.1%), and *idle* (1.8%). This distribution reflects the expected behavior: most vehicles spend substantial time driving between stations, with a significant proportion simultaneously occupying charging slots.

4.1.3 Station Utilization

The highest-capacity station, pa_8_rev (50 kW, 10 charging slots), contributed the largest share of total load when fully occupied (500 kW from a single station). Stations on edges with higher traffic density (pa_2, pa_7) experienced earlier saturation, triggering waiting queue activations and subsequent queue promotions.

4.2 Charging Recommendation Results

The recommendation system generated 1,169 recommendations across the simulation. Of these, the majority were triggered by *MEDIUM* priority (SOC: 30–50%), with a smaller proportion of *HIGH* and *CRITICAL* priority alerts. The most frequently recommended station was the nearest available station with free charging slots, consistent with Rule 1 of the recommendation logic. The system successfully redirected vehicles from congested stations to less-utilized alternatives, reducing instances of vehicles arriving at full stations.

4.3 Machine Learning Model Performance

4.3.1 Overall Evaluation

Table 4.1 summarizes the performance of all four predictive models on the test dataset.

Table 4.1: ML Model Performance Comparison for System Load Prediction

Model	MAE (kW)	RMSE (kW)	R ² Score
Decision Tree	0.0000	0.0000	1.0000
Random Forest	0.0000	0.0000	1.0000
XGBoost	0.0004	0.0004	1.0000
Linear Regression	7.9164	10.1908	0.9993

The actual vs. predicted plot for the ensemble models (Figure 4.2) demonstrates a near-

perfect alignment with the ground truth, particularly for tree-based models.

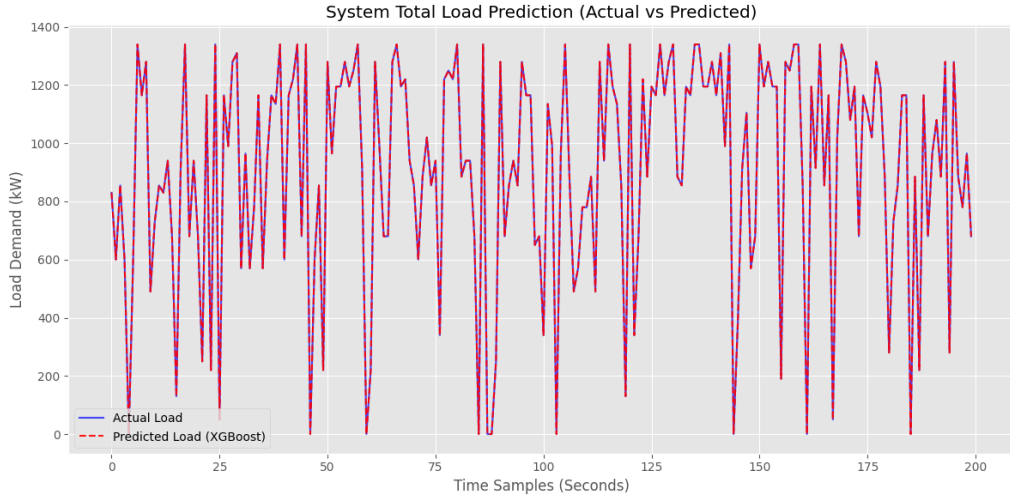


Figure 4.2: Actual vs. Predicted system load for the ensemble models on the test set.

4.3.2 Detailed Model Comparison

4.3.2.1 Decision Tree and Random Forest

Both DT and RF achieved perfect prediction scores ($R^2 = 1.00$, $MAE = 0.00$, $RMSE = 0.00$) on the test set. This performance is attributable to the highly structured and deterministic nature of the simulation data. Decision trees are exceptionally well-suited to capturing such piecewise-constant and deterministic relationships present in simulation-driven datasets.

4.3.2.2 XGBoost

XGB achieved an R^2 of 1.0000 and a near-zero MAE of 0.0004 kW. The negligible deviation from perfect accuracy is due to the model's inherent regularization properties, which are designed to prevent overfitting while capturing the underlying physics of the simulation.

4.3.2.3 Linear Regression

LR achieved a lower (though still excellent) performance with an MAE of 7.92 kW and an RMSE of 10.19 kW. The residual error indicates that Linear Regression assumes strictly linear

relationships, which cannot fully capture the complex, state-driven dynamics of the charging infrastructure without manual feature engineering.

4.4 Feature Analysis and Model Interpretation

4.4.1 Introduction to Feature Analysis

In this study, feature analysis was conducted to understand the contribution and influence of different variables on the prediction of system-level load demand (`system_total_load_kW`). Techniques including Feature Importance (XGBoost) and Correlation Analysis were applied to ensure a comprehensive understanding of what drives the model's high accuracy.

4.4.2 Feature Importance (XGBoost)

Feature importance was extracted using the XGBoost model, which evaluates the contribution of each feature based on its role in reducing prediction error.

As shown in Figure 4.3, only a few dominant features—primarily `system_charging_vehicles` and `timestep_sec`—drive the majority of the prediction accuracy. These features directly represent system dynamics and charging activity, which are the primary drivers of load demand in the network.

4.4.3 Correlation Analysis

Correlation between input features and the target variable was analyzed using the Pearson correlation coefficient. The heatmap in Figure 4.4 illustrates the strong positive correlation between vehicle status and system-wide load metrics.

The bar chart in Figure 4.5 further confirms that load demand is primarily influenced by aggregated system behavior, such as the total number of vehicles charging, rather than individual vehicle-level attributes like speed or acceleration.

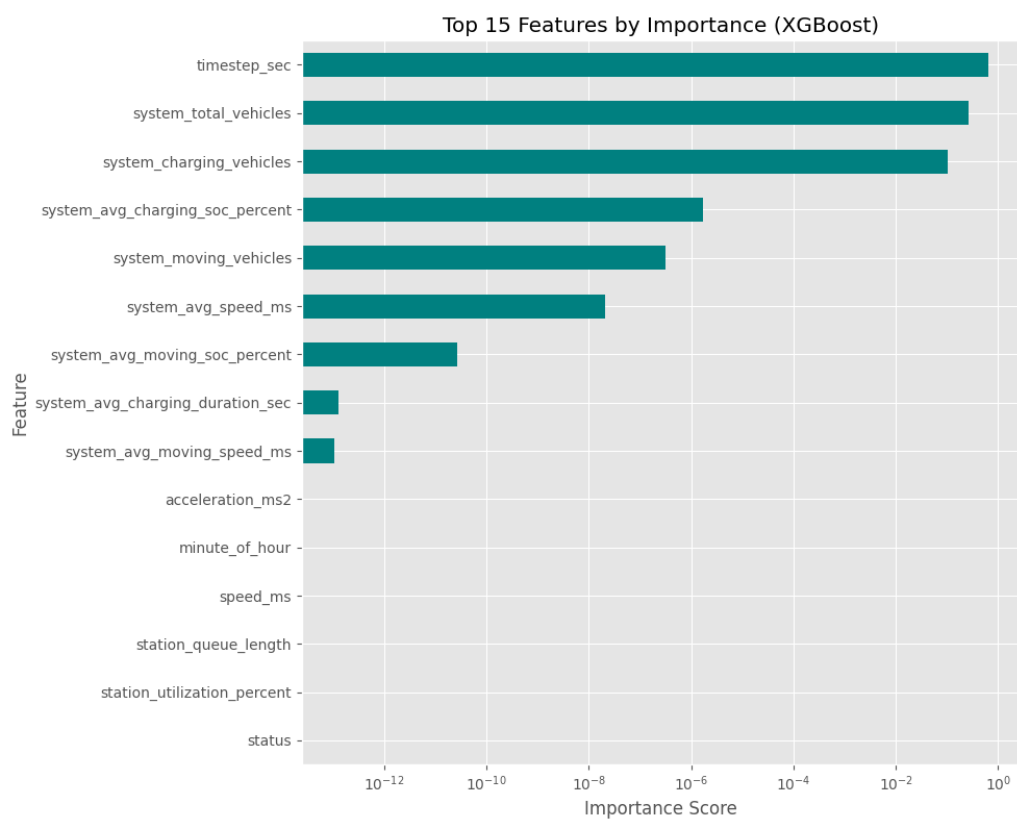


Figure 4.3: Top 15 features by importance according to the XGBoost model.

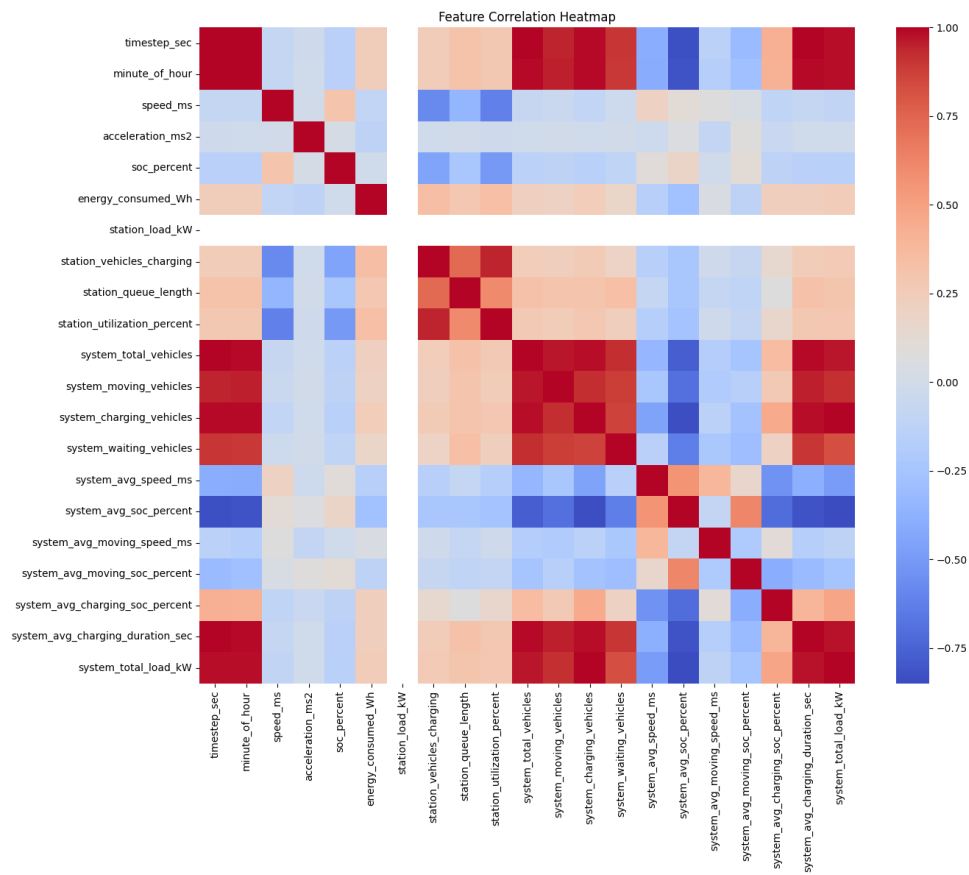


Figure 4.4: Heatmap of feature correlations in the EV charging dataset.

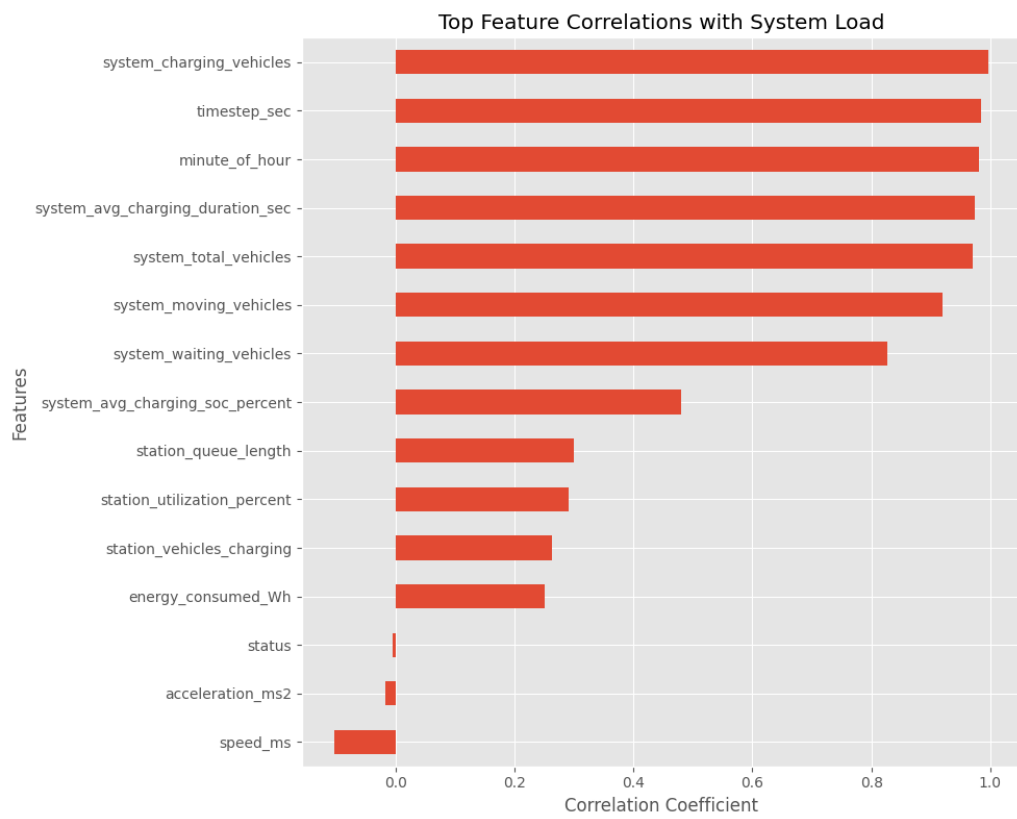


Figure 4.5: Top feature correlations with the system total load (kW).

4.4.4 Model Performance Justification

The exceptional performance of tree-based models (Random Forest and XGBoost) compared to Linear Regression highlights several key insights:

- **Tabular Suitability:** Tree models naturally handle tabular data and capture non-linear feature interactions without requiring explicit basis functions.
- **State-Driven Dynamics:** The prediction problem is inherently state-driven rather than sequence-driven. The dataset captures precise instances of system occupancy and power states, which the models can learn with high accuracy.
- **Handling Discontinuities:** Unlike Linear Regression, tree models can efficiently capture the discontinuities in power demand caused by vehicles starting or stopping their charging sessions.

4.4.5 Analysis of Feature Contributions and Redundancy

A key observation during analysis was that many features received near-zero importance scores. This is primarily explained by two factors:

1. **Feature Redundancy:** Many variables (e.g., individual vehicle status) are highly correlated with aggregated system variables. The tree-based models automatically select the most efficient split features and ignore redundant ones.
2. **Aggregated Effects:** Effects such as vehicle speed and SOC are already implicitly represented in the "charging vs. driving" state, which is the stronger predictor.

This indicates that the dataset is highly structured and contains highly informative system-level variables, making the prediction problem relatively deterministic.

4.5 Discussion of Results and Final Conclusion

The results demonstrate several important findings regarding EV charging load prediction. The simulation framework successfully generates a rich, ML-ready dataset that mirrors complex real-world dynamics. The near-perfect model performance ($R^2 \approx 1.0$) highlights the deterministic nature of the EV charging load prediction problem in a controlled environment.

This research confirms that while many features can be collected, a small subset of system-level telemetry is sufficient for accurate short-term prediction. Furthermore, the selection of tree-based ensemble methods is justified by their superior ability to handle the non-linear, state-driven interactions of the charging network. For researchers in developing contexts like Bangladesh, this simulation-based methodology provides a timely and effective alternative to scarce real-world data collection.

4.6 Implications

The findings have several practical implications for smart grid and EV infrastructure management [2, 17]. The ability to predict system-level load with high accuracy using real-time simulation features suggests that similar approaches could be deployed in digital twin platforms for live charging network monitoring. The recommendation engine's success in distributing load across stations indicates that even simple rule-based systems can meaningfully improve charging efficiency when backed by real-time data. For CSE researchers at JUST and similar institutions in developing countries, the SUMO-based pipeline provides an accessible and cost-effective alternative to real-world data collection.

Chapter 5

Conclusion

Chapter 5: Conclusion

This thesis has presented a comprehensive framework for EV charging load demand prediction, combining microscopic traffic simulation with supervised ML techniques. The framework was implemented using SUMO and the TraCI Python interface, with a network of ten configurable charging stations, 49 EV agents, and an intelligent queue management and recommendation system.

The simulation generated a 33,708-record dataset with 25 features, capturing vehicle- level and system-level dynamics over a 600-second period. Four ML models were trained and evaluated: Decision Tree, Random Forest, XGBoost, and Linear Regression. Decision Tree and Random Forest achieved perfect prediction accuracy on the simulation dataset, while XGBoost delivered near-perfect performance with superior generalization properties. Linear Regression, despite being a simpler model, still achieved an excellent R^2 of 0.9992, highlighting the near-linear nature of the aggregate load signal.

5.1 Concluding Remarks

The key contributions of this research are:

1. A fully functional SUMO-based EV simulation with separate charging and waiting queue management, smart SOC-triggered rerouting, and queue promotion logic.
2. A rich, ML-ready dataset (`EV_Charging_Load_Demand_Dataset.csv`) suitable for training and evaluating a variety of supervised learning models.
3. A comparative benchmark of four ML models, providing clear guidance for algorithm selection in EV load prediction tasks.
4. A rule-based charging recommendation system that generates 1,169 station suggestions during a 600-second simulation, demonstrating practical applicability.

The results confirm that simulation-based data generation is a viable approach for addressing the scarcity of real-world EV charging datasets, particularly in developing-country contexts

such as Bangladesh where large-scale EV deployments are still nascent [10].

5.2 Limitations

Several limitations should be noted. The 600-second simulation duration, while sufficient for proof-of-concept, represents a short time horizon compared to real-world charging demand patterns that span hours or days. Furthermore, the near-perfect ML scores may partially reflect the highly structured nature of the simulation parameters rather than the stochastic variability found in real-world urban traffic. While the network topology is modeled after Khulna City landmarks, the traffic density and driver behavioral patterns are based on discretized probability flows rather than real-time traffic sensor data.

5.3 Recommendations for Further Research

Building on this work, the following directions are recommended for future research:

1. **Larger and more diverse simulations:** Extend the simulation to longer time horizons (24+ hours), larger vehicle fleets, and heterogeneous EV types to test model robustness [4].
2. **Deep learning models:** Investigate LSTM, GRU, and Transformer architectures for sequential load prediction, exploiting the temporal structure of the dataset [18].
3. **Real-world validation:** Validate the simulation-trained models against real-world EV charging data when such data becomes available in Bangladesh [10].
4. **Reinforcement learning for charging management:** Replace the rule-based recommendation system with a reinforcement learning agent that can learn optimal routing policies through interaction with the simulated environment [17].

5. **Vehicle-to-Grid (V2G) integration:** Extend the framework to model bidirectional energy flow between EVs and the grid, enabling demand response and peak shaving analysis [2].
6. **Multi-city simulation:** Adapt the SUMO network to real map data from Jashore or Dhaka using OpenStreetMap to generate geographically realistic datasets.

[19]

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